Cognitive modeling meets computational linguistics

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Who am I?

Yevgen Matusevych /jɛwˈgɛn mɐtuˈsɛʊɪtʃ/

since 2023	assistant professor, Computational Linguistics, Groningen
2018-23	postdoc, NLP & Center for Language Evolution, Edinburgh
2017-18	postdoc, Computational Linguistics, Toronto
2012-16	PhD student, Tilburg

Who am I?

Yevgen Matusevych /jɛwˈɡɛn mɐtuˈsɛʊɪtʃ/

- Cognitive models of language learning and processing
- Comparing engineering models (LMs, speech models) to humans
- Across linguistic domains: phonetics, semantics, syntax, etc.
- Special interests: categorization, multilingualism

Who are you?

Let’s do a quick round!

- Your name
- Your university
- Your study program
- Your research interests (in 1-2 sentences)
- (Optional) Why this course?
- (Optional) Your experiences with cognitive modeling

What is this course?

Mon	What makes a computational model cognitive?
Tue	Architectural and data choices
Wed	Learning representations
Thu	Multilingual learning
Fri	Modeling across domains and modalities

What kind of modeling?

Computational cognitive modeling traditions / communities:

- Symbolic architectures: ACT-R, EPIC
- ‘Rational’ models: Rational Speech Act, Bayesian learning
- Mechanistic processing models: surprisal, lexical competition
- Neural / representation learning: LMs, speech, multimodal

Why representation learning?

Modern architectures actually **work**

- They do very well in NLP / speech processing tasks
- They learn *something* useful
- But: is that *something* also helpful for cognitive science / linguistics?

Day 1
What makes a computational model cognitive?

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A historic example

English past tense - mainstream view in the 1980s:

Children learn regular past tense by acquiring a symbolic rule; irregular forms are exceptional items stored separately.

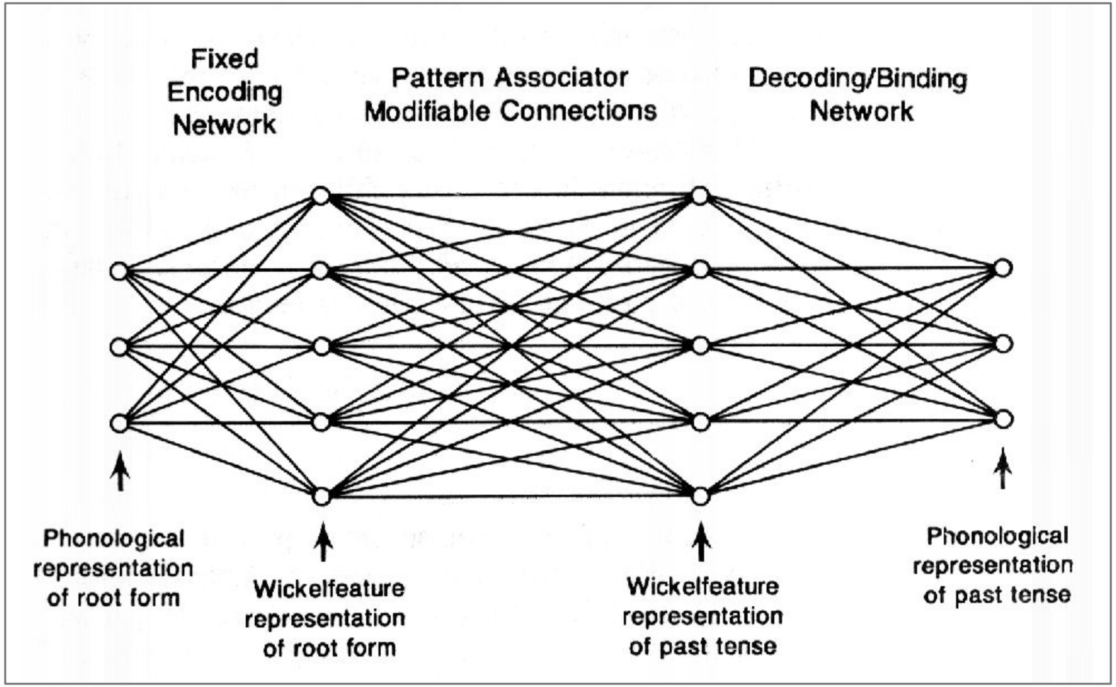
A historic example

Rumelhart & McClelland (1986): a neural network model for learning English past tense inflection

Claim: *An associative mechanism without explicit rules can learn both regular and irregular English past tense forms*

Therefore: *Explicit rules need not be stored as such*

Rumelhart & McClelland (1986)



encoder - pattern associator - decoder

Rumelhart & McClelland (1986)

TABLE 6

THE SIXTEEN WICKELFEATURES FOR THE WICKELPHONE $_kA_m$

Feature	Preceding Context	Central Phoneme	Following Context
1	Interrupted	Vowel	Interrupted
2	Back	Vowel	Front
3	Stop	Vowel	Nasal
4	Unvoiced	Vowel	Voiced
5	Interrupted	Front	Vowel
6	Back	Front	Front
7	Stop	Front	Nasal
8	Unvoiced	Front	Voiced
9	Interrupted	Low	Interrupted
10	Back	Low	Front
11	Stop	Low	Nasal
12	Unvoiced	Low	Voiced
13	Interrupted	Long	Vowel
14	Back	Long	Front
15	Stop	Long	Nasal
16	Unvoiced	Long	Voiced

Wickelfeatures

Rumelhart & McClelland (1986)

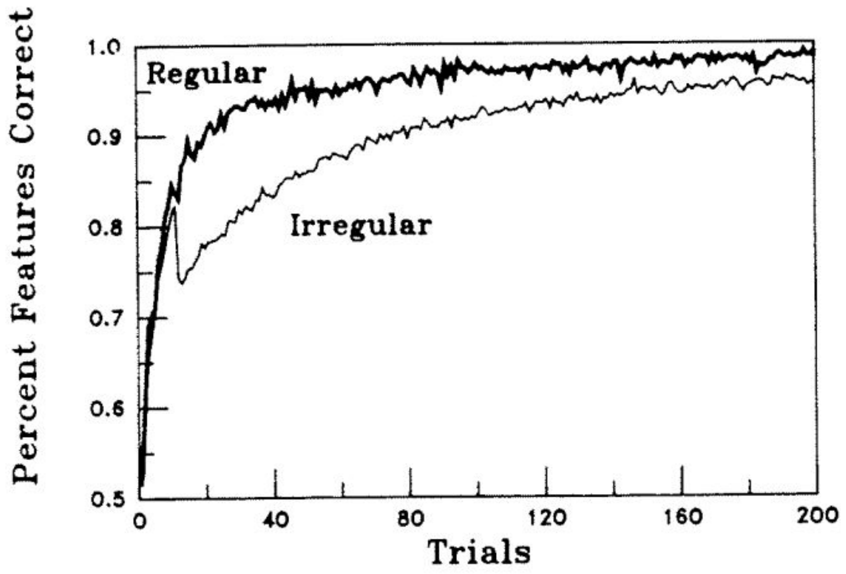


FIGURE 4. The percentage of correct features for regular and irregular high-frequency verbs as a function of trials.

The U-shaped curve

A historic example

Pinker & Prince (1988): a 120-page critique

- The model does not eliminate the possibility of having rules
- Wickelfeatures are implausible
- Model produces unrealistic errors (unparseable feature sets)
- U-shaped learning curve is a learning schedule artifact
- The model only does phonological pattern matching

Cognitive plausibility dimensions

Plausibility dimensions

Beinborn & Hollenstein (2024):

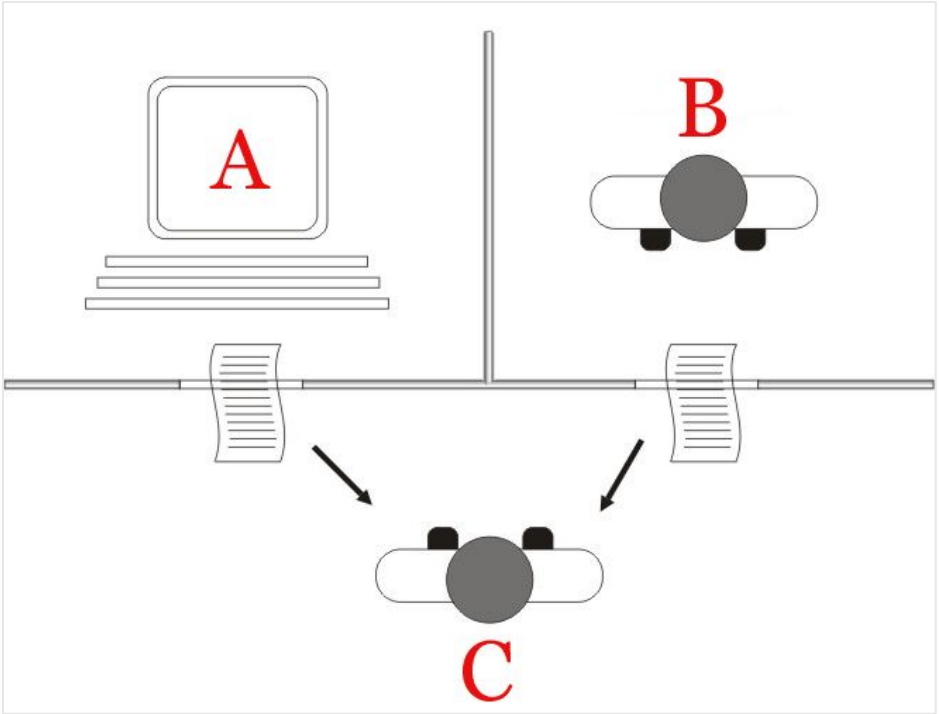
- **Behavioral:** does a model behave the way humans do?
- **Representational:** representations similar to human ones?
- **Procedural:** strategies similar to human ones?

More specific points:

- **Developmental:** is the learning trajectory similar to human?
- **Input:** can a model learn from input similar to human?

Behavioral plausibility

Does the model "behave" the way humans do?



Turing test, from [Wikipedia]

Behavioral plausibility

How do we test model behavior?

- Testing linguistic phenomena
  - Targeting a specific behavior
  - Minimal-pair tasks - e.g., Linzen et al., 2016:
    The keys to the cabinet is/are …
  - Standardized benchmarks - e.g., BLiMP (Warstadt et al., 2020):
    a number of *paradigms* targeting various linguistic phenomena

Behavioral plausibility

Phenomenon	N	Acceptable Example	Unacceptable Example
ANAPHOR AGR.	2	*Many girls insulted themselves.*	*Many girls insulted herself.*
ARG. STRUCTURE	9	*Rose wasn't disturbing Mark.*	*Rose wasn't boasting Mark.*
BINDING	7	*Carlos said that Lori helped him.*	*Carlos said that Lori helped himself.*
CONTROL/RAISING	5	*There was bound to be a fish escaping.*	*There was unable to be a fish escaping.*
DET.-NOUN AGR.	8	*Rachelle had bought that chair.*	*Rachelle had bought that chairs.*
ELLIPSIS	2	*Anne's doctor cleans one important book and Stacey cleans a few.*	*Anne's doctor cleans one book and Stacey cleans a few important.*
FILLER-GAP	7	*Brett knew what many waiters find.*	*Brett knew that many waiters find.*
IRREGULAR FORMS	2	*Aaron broke the unicycle.*	*Aaron broken the unicycle.*
ISLAND EFFECTS	8	*Whose hat should Tonya wear?*	*Whose should Tonya wear hat?*
NPI LICENSING	7	*The truck has clearly tipped over.*	*The truck has ever tipped over.*
QUANTIFIERS	4	*No boy knew fewer than six guys.*	*No boy knew at most six guys.*
SUBJECT-VERB AGR.	6	*These casseroles disgust Kayla.*	*These casseroles disgusts Kayla.*

Table 2: Minimal pairs from each of the twelve linguistic phenomenon categories covered by BLiMP. Differences are underlined. *N* is the number of 1,000-example minimal pair paradigms within each broad category.

Warstadt et al. (2020)

Behavioral plausibility

How do we test model behavior?

- Testing robustness and generalizability
  - Test data perturbations - e.g., semantically anomalous sentences: *The ideas rowing in my lamp's economy sleep.* (Gulordava et al., 2018)
  - Training data manipulations - e.g., removing a construction
  - Multilinguality - e.g., MultiBLiMP (Jumelet et al., 2026)
  - Comparing error patterns - e.g., error direction
  - Trajectories rather than single points

Behavioral plausibility

- A simple version: Does the model produce human-like qualitative patterns?
- A stronger question: Does the model produce human-like patterns item-by-item?

Behavioral plausibility: Surprisal

Example: surprisal theory

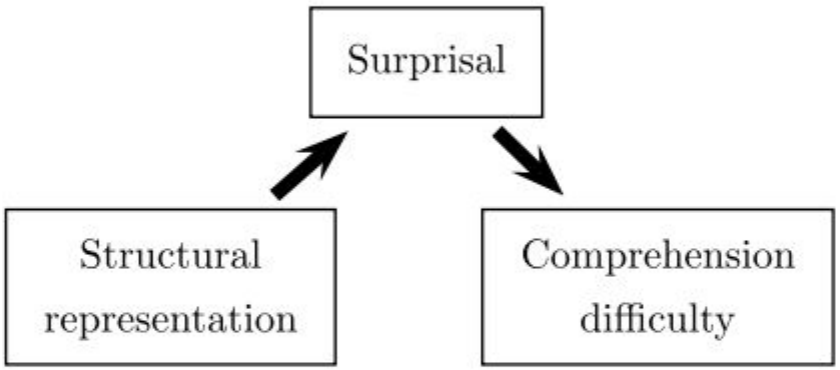
$$\text{surprisal}(w) = -\log P(w \mid \text{context})$$

Hale (2001):

"The proposal is that a person's difficulty perceiving syntactic structure be modeled by word-to-word surprisal (Attneave, 1959, page 6) which can be directly computed from a probabilistic phrase structure grammar."

Behavioral plausibility: Surprisal

“surprisal serves as a causal bottleneck between the linguistic representations constructed during sentence comprehension and the processing difficulty incurred at a given word within a sentence”



(b) Surprisal as a causal bottleneck mediating effect of representation on processing

Levy (2008)

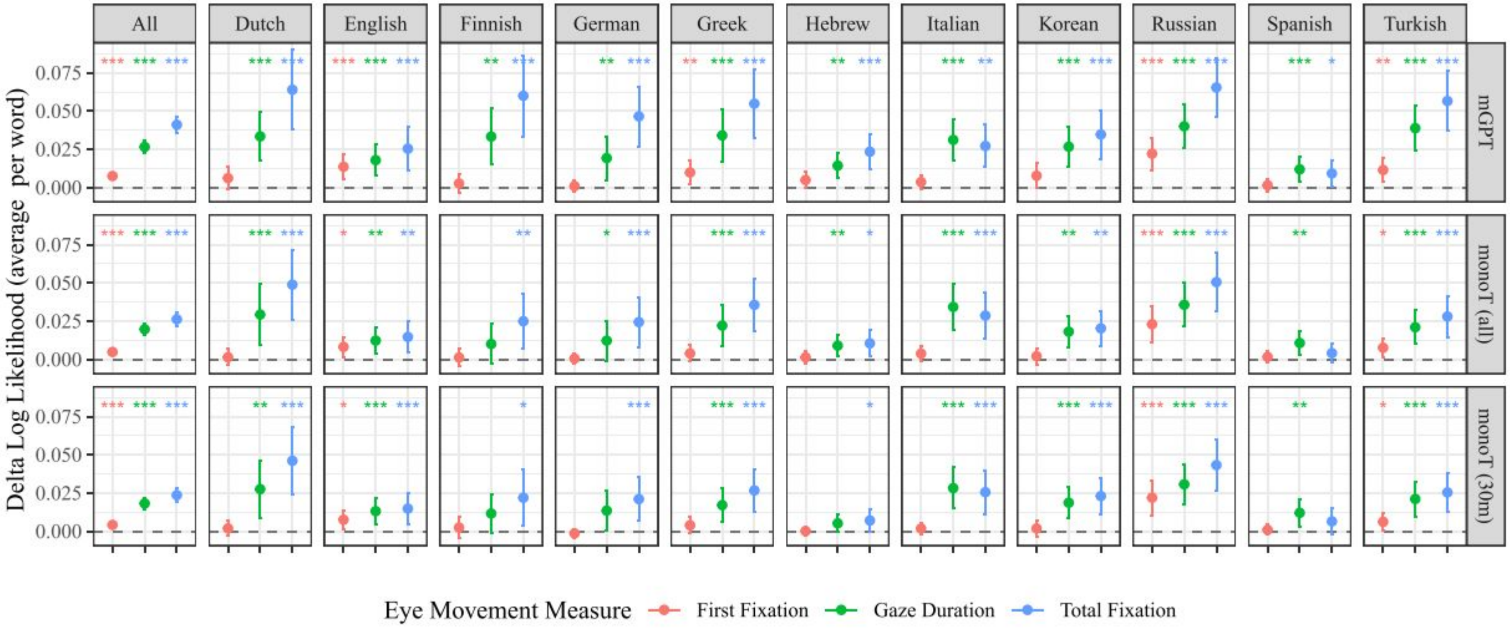


Figure 1: **Predictive Power of Surprisal Across Languages:** Positive values mean surprisal contributes to predicting the reading times over a baseline where surprisal is removed. Error bars indicate 95% confidence intervals. Asterisks indicate the significance of a paired permutation test. We find a consistent significant effect of surprisal across languages for language models that are both multilingual (top row) and monolingual (bottom two rows), and for both progressive gaze duration and total fixation.

Wilcox et al. (2023)

Behavioral plausibility: Surprisal

- Processing difficulty $\propto$ surprisal (Hale, 2001; Levy, 2008)
- Human data: reading time / eye-tracking / N400
- Model probabilities: an LM's next-word distribution

- Once we've established that surprisal is a good linking hypothesis, we can use it to compare model output distributions to human behavior - i.e., testing behavioral plausibility

Behavioral plausibility: Testing feasibility

Winther, Matusevych & Pickering (2021), *Cumulative frequency can explain cognate facilitation in language models*

- Cognates (shared form+meaning across languages) are processed faster by bilinguals than non-cognates
- An LM simulating bilingual exposure reproduces the cognate facilitation effect seen in human data

Behavioral plausibility: Summing up

- We can discover a model’s input-output patterns
- We can infer strategies a model might have learned
- We (usually) *cannot* say with certainty why a behavior is observed

- Extreme case: Searle's Chinese Room (1980)

Behavioral plausibility: Diving deeper

Which items are easy or hard?

What kinds of errors occur?

Does model uncertainty track human difficulty?

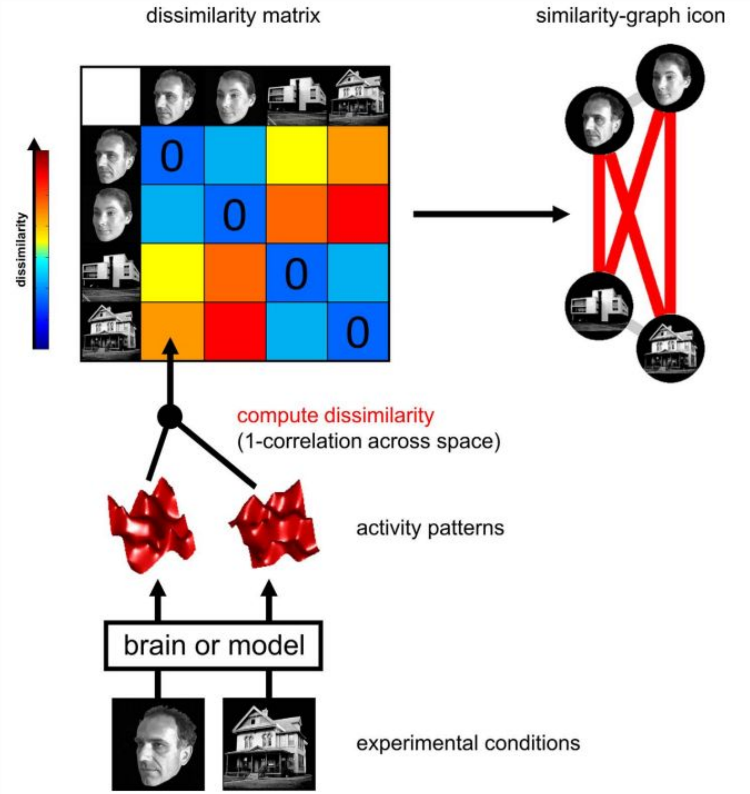
Are errors distributed across subgroups in human-like ways?

Does the model generalize beyond the benchmark?

Does it learn easier phenomena before harder ones?

Representational plausibility

Does the model encode and organize information the way humans do?



Kriegeskorte et al. (2008)

Representational plausibility

How do we compare models' and humans' representations?

Two issues:

- Human representations are often defined via behavior
- Large neural models' representations are difficult to examine

(Classes of) approaches:

- Look for linguistic knowledge in models' representations
- Compare models' representations to brain activation patterns

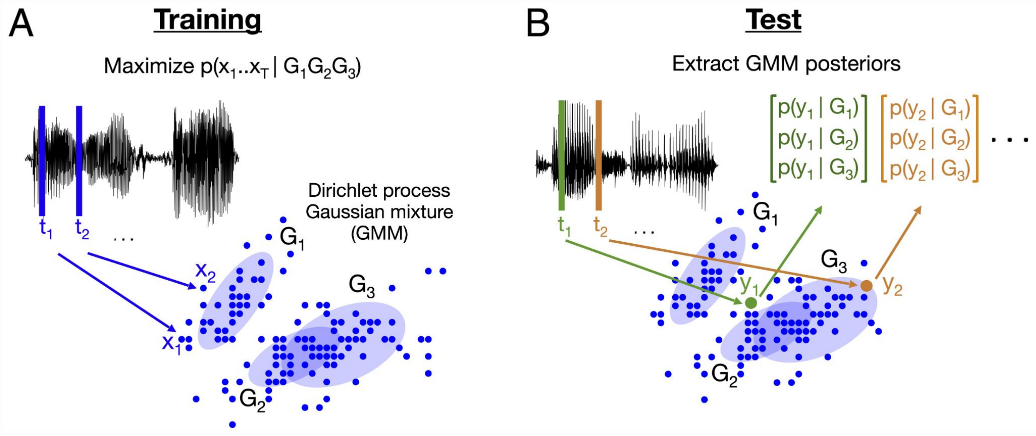
Representational plausibility

- Looking for linguistic knowledge – using a shortcut: implicitly comparing models' *representations* to human *behavior*

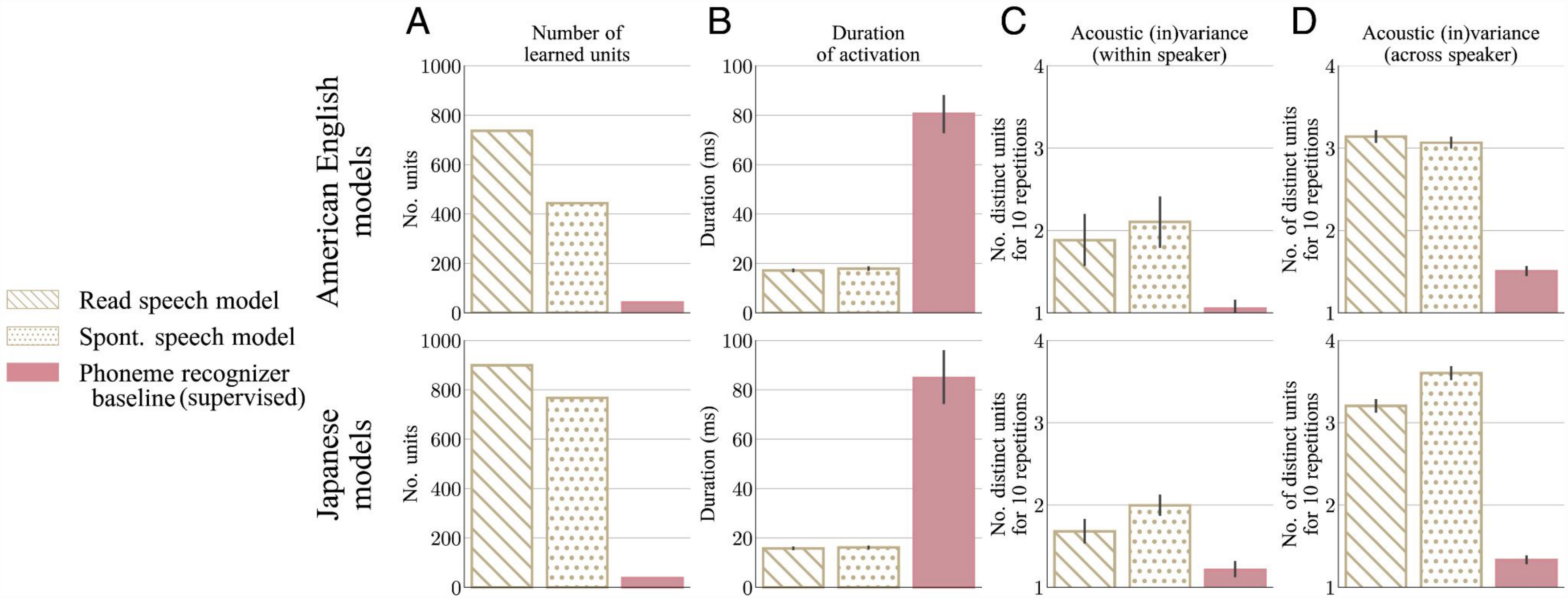
- Example: compare distances in vector space to human judgments (e.g., Hill et al., 2015: SimLex999)

Representational plausibility

A probabilistic model trained on raw English vs. Japanese speech



Schatz et al. (2021)



Schatz et al. (2021)

Representational plausibility

One method commonly used in ~2017-2020: **probing classifiers** (see Belinkov, 2022)

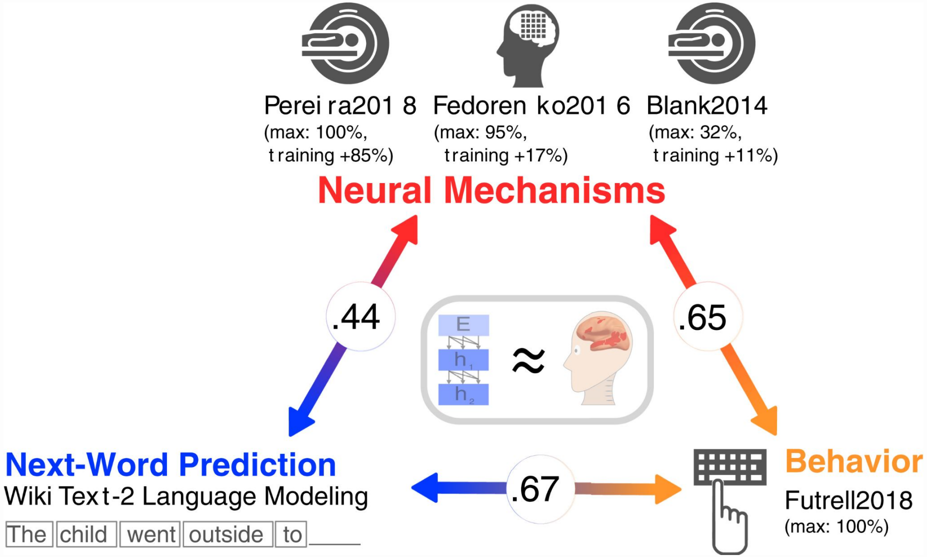
Idea: train a small supervised classifier to predict a linguistic property from its activations in the model

Prediction accuracy ≈ the extent to which the property is encoded

Representational plausibility

Using brain imaging - e.g., Schrimpf et al. (2021):

- Transformers predict nearly all explainable variance in neural responses to sentences
- Generalize across datasets and modalities (functional MRI and electrocorticography).



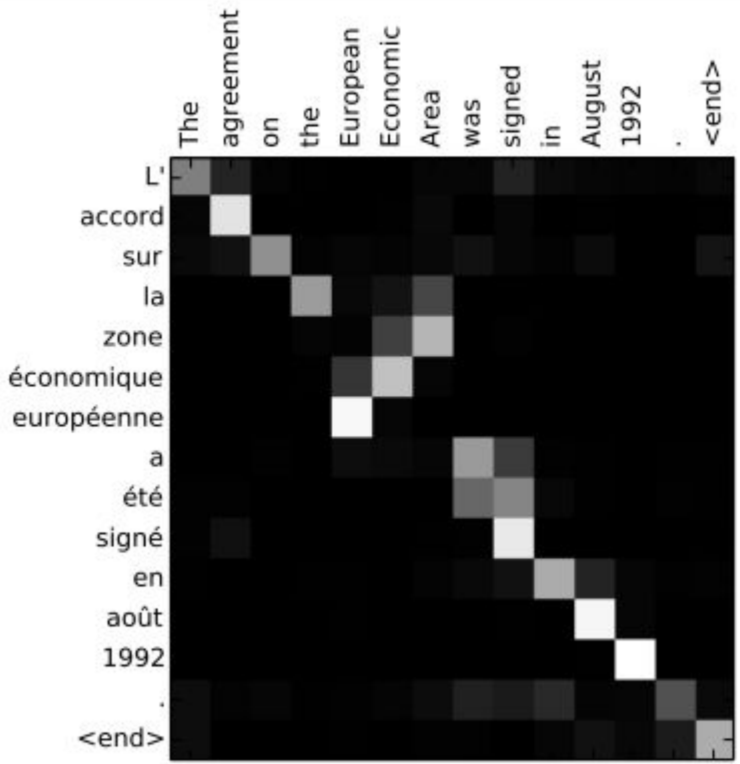
Schrimpf et al. (2021)

Representational plausibility

- Representational evidence is stronger than behavior alone
- But a property may be contained in a representation without being used by the model
- Classic example: probing classifiers (Ravichander et al., 2021, Elazar et al., 2021)

Procedural plausibility

How exactly does the model use information over time to produce a certain behavior?



Bahdanau et al. (2014)

Procedural plausibility

Autoregressive vs masked language modeling:

- Autoregressive: left-to-right processing, incremental prediction
- Masked: additional right context, can capture bidirectional integration in human processing

Procedural plausibility

Hollenstein & Beinborn (2021)

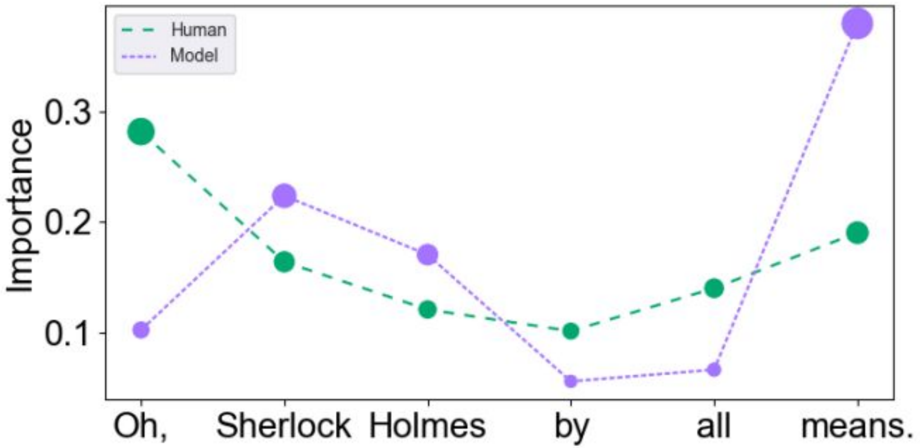


Figure 3: Relative importance values for an example sentence from the GECO corpus for the BERT model and the human values.

Procedural plausibility

- “Procedural” computations are usually grounded in interpretability methods
- If the interpretability tools are not robust, neither are “procedural” conclusions (e.g., Jain & Wallace, 2019)

Representational/procedural plausibility

Model interpretability (beyond probing classifiers):

- Causal interventions - e.g., amnesic probing (Elazar et al., 2021)
- Circuit analysis - reverse-engineering model behavior (e.g., Wang et al., 2023)
- Sparse autoencoders - replacing neurons with more meaningful features (Bricken et al., 2023)
- More on model interpretability: https://projects.illc.uva.nl/indeep/tools-and-tutorials/

Measures

- Offline (acceptability judgments, lexical decisions, ratings, etc.)
- Online (reading times, eye-tracking measures)
- Brain (EEG, fMRI)

One important thing often overlooked: *A cognitive claim is only as strong as the human evidence it is linked to.*

Cognitive plausibility as a lens

- Cognitive plausibility is not a simple binary construct
- A better framing: Can a computational model test a specific claim about human cognition?
- Factors to consider:
  - Behavioral / representational / procedural dimension?
  - Is it tested on strong evidence?
  - Is the testing robust?
  - Is the result causal or correlational?

Model audit

- What is the cognitive target?
  - past tense? reading time? phonetic discrimination?
- What is the model target?
  - classification? prediction? reconstruction? contrastive learning?
- Which dimension is being tested?
  - behavior? representation? procedure?
- What human evidence is used?
  - adult behavior? developmental data? reading times?
- What would count as a failure?
  - wrong errors? wrong learning curve? wrong representation?
- What alternative explanation remains?
  - frequency? dataset artifact? architecture? evaluation setup?

A historic example

Plunkett & Marchman (1991, 1993): multilayer networks that produced robust U-shaped over-regularization

Marcus et al. (1995): German -s plural behaves as a productive default despite being a minority form

McClelland & Patterson (2002) vs. **Pinker & Ullman (2002)**: an exchange in *Trends in Cognitive Sciences*; Ullman's declarative/procedural model vs. McClelland & Patterson's single-mechanism reply

A historic example connecting to modernity

Albright & Hayes (2003): a new model, Minimal Generalization Learner, induces many stochastic rules at varying levels of generality

Kirov & Cotterell (2018): a character-level encoder-decoder model, argued that they overcome Pinker & Prince's objections

Corkery, Matusevych & Goldwater (2019): tested the model more thoroughly and showed that the model doesn't behave the way humans do (and is unstable)

Character-level models

- Kirov & Cotterell (2018) use a modern NN architecture: character-level encoder-decoder model

- It arguably overcomes most criticisms of Pinker & Prince (1988)

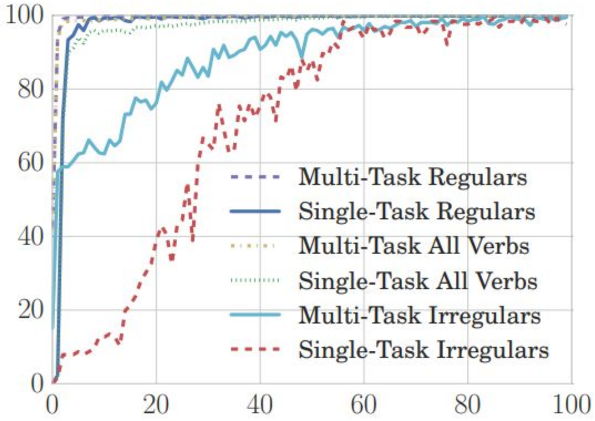


Figure 1: Single-task vs. multi-task. Learning curves for the English past tense. The x-axis is the number of epochs (one complete pass over the training data) and the y-axis is the accuracy on the training data (not the metric of optimization).

Kirov & Cotterell (2018)

Character-level models: instability

Corkery et al. (2019) tested the same model more thoroughly:

- Multiple random initializations

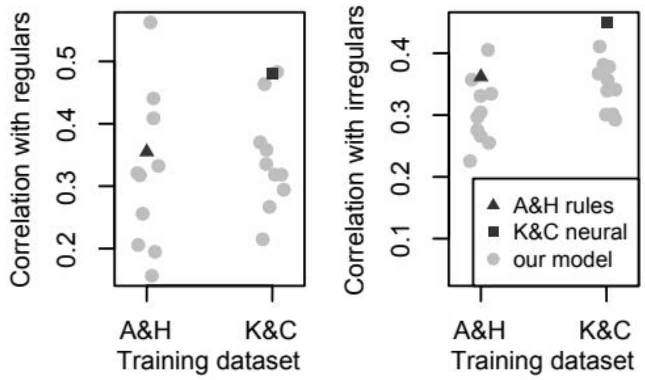


Figure 2: Spearman correlation coefficients between model scores and human production probabilities, using the A&H and K&C training data. Values reported by K&C and A&H are shown in addition to those of our models. Horizontal jitter is added for readability.

Corkery et al. (2019)

Character-level models: second choices

Corkery et al. (2019) tested the same model more thoroughly:

- Multiple random initializations
- Second most preferred form

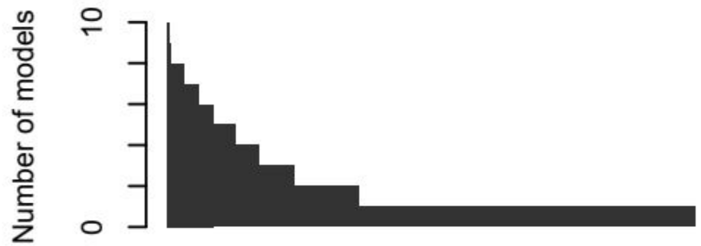


Figure 4: The number of models (of the 10 trained on the A&H dataset) which agree on the second-place past tense form. The X-axis shows 281 different past tense forms (for 59 nonce words in the present tense), and the Y-axis shows, for each form, how many times a model places it in the second position in the beam.

Corkery et al. (2019)

Character-level models: aggregate model

Corkery et al. (2019) tested the same model more thoroughly:

- Multiple random initializations
- Second most preferred form
- An aggregate model (sampling)

Character-level models: aggregate model

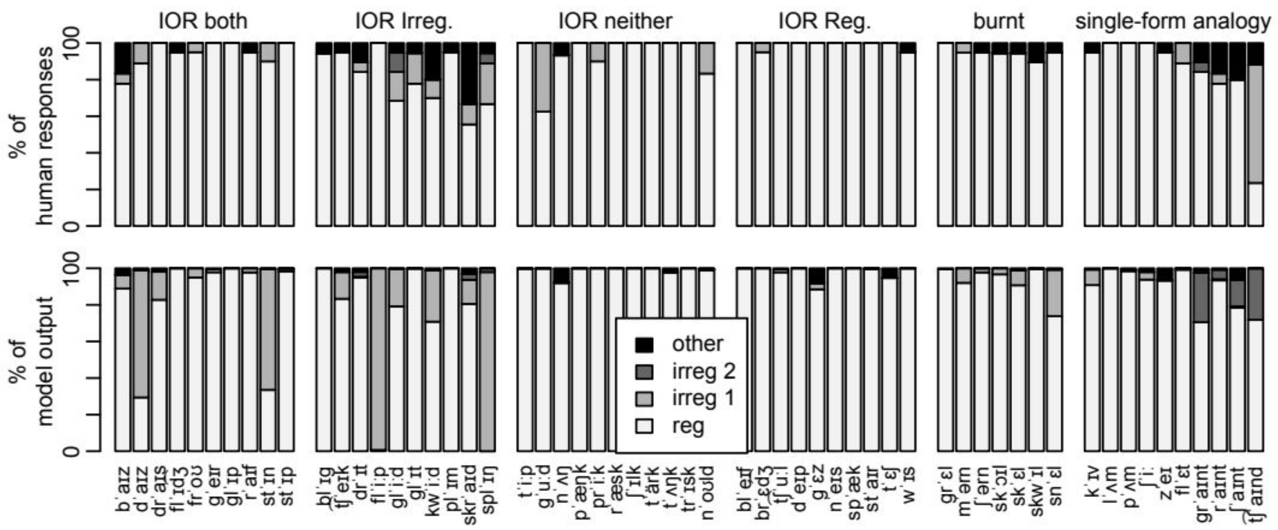


Figure 5: Percentage of regular, irregular, and "other" responses produced by humans (top) and the model (bottom). Each of the six blocks corresponds to a different category of nonce words (see Section 2.2).

Corkery et al. (2019)

Character-level models: bottom line

Corkery et al. (2019):

- The ED model learns past tense forms of verbs in the training data
- Behaviour on nonce verbs correlates poorly with human experimental data (worse than A&H’s rule-based model)

A historic example connecting to modernity

- One possible question: Who is right in this debate?
- Better question: Which statement(s) are we making?
    - Strong nativism (“NNs cannot”)? Not supported.
    - Strong eliminativism (“no rules”)? Not supported.
    - Dual-mechanism vs. graded generalization? Unresolved.

What makes a model cognitive?

- Intrinsic properties matter a lot, but only with respect to the specific claim about humans
- The same architecture trained on the same data can be either an engineering artifact or a model testing a claim about cognition

Tomorrow: *architectural and data choices*

Thanks!